Bird Abundance

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Ideas for Analysis

Topic:

Overall variation in bird species abundance by season based on water elevation, and variation in species abundance by season between taxonomic groups.

Question(s):

- How does water elevation effect bird species abundance within the Waterfowl and Shorebird taxonomic groups across seasons?

Addressing Rubric Feedback: focusing on one, more specific, research question as opposed to three similar questions.

Ideas for analysis and backround

- Plot showing total shorebird counts by species in 2025 water year
- Plot showing total waterfowl counts by species in 2025 water year
- Plot showing trends in water elevation by date in 2025 water year
- Plot showing waterfowl and shorebird species counts across water elevations in 2025 water year

Background

Visualizations

1. Set up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

#Wrap Axis Label
library(scales)

#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
library(flextable)
library(ggridges)
library(NatParksPalettes)
library(paletteer)
```

Data

```
birds <- read_csv(here("data", "birds.csv"))

venoco_water <- read_csv(here("data", "venoco-water.csv"))
```

2. Cleaning

Wrangling birds

```r
#created a new object from birds
bird_wy2025 <- birds %>%
  # exclude repeat observations
  mutate(observation_date = as.Date(observation_date, format = "%m/%d/%y"))
   ↪  %>%
  filter(observation_date > as.Date("2024-11-13") & observation_date <
   ↪  as.Date("2025-10-22"),
         repeat_observation == "No",
         e_bird_group %in% c("Waterfowl",
                             "New World Sparrows",
                             "Martins and Swallows",
                             "Shorebirds")) %>%
  # filling in taxanomic information about order and families for eBird
   ↪  groups
  mutate(lowest_group = case_when(
    e_bird_group == "New World Sparrows" ~ "Passerellidae",
    e_bird_group == "Martins and Swallows" ~ "Hirundinidae",
    e_bird_group == "Shorebirds" ~ "Charadriiformes",
    e_bird_group == "Waterfowl" ~ "Anatidae"
  ))



# creating new object from most_abundant_groups
abundant_species_counts <- bird_wy2025 %>%
  # grouping by speices
  group_by(season, e_bird_group, species) %>%
  # calculating total counts for each species
  summarize(total_count = sum(count)) %>%
  # arranging in descending order
  arrange(-total_count)
```

Wrangling Water

```r
# Creating a new object from venoco_water
venoco_clean <- venoco_water |>
```

```
  # filter to only include 2025 water year
  filter(wy == "2025") |>
  # creating a new column to convert water elevation in ft to meters
  mutate(comp_level_m = (comp_level_ft + 2.96) * 0.3048) |>
  # selecting columns of interest
  select(date, comp_level_m, temperature)


# create a new object from Venoco_clean
venoco_daily <- venoco_clean |>
# group by date
  group_by(date) |>
# calculate mean water level for each day
  summarize(mean_level_m = mean(comp_level_m, na.rm = TRUE)) |>
# ungroup the data frame
  ungroup()
```

Shorebird Wrangling

```
shorebird_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Charadriiformes") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  #arrange in descending order by total counts
  arrange(-total_count)
```

Waterfowl Wrangling

```
waterfowl_total_count <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(lowest_group == "Anatidae") |>
  #group by species
  group_by(species) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
```

```
  #arrange in descending order by total counts
  arrange(-total_count)
```

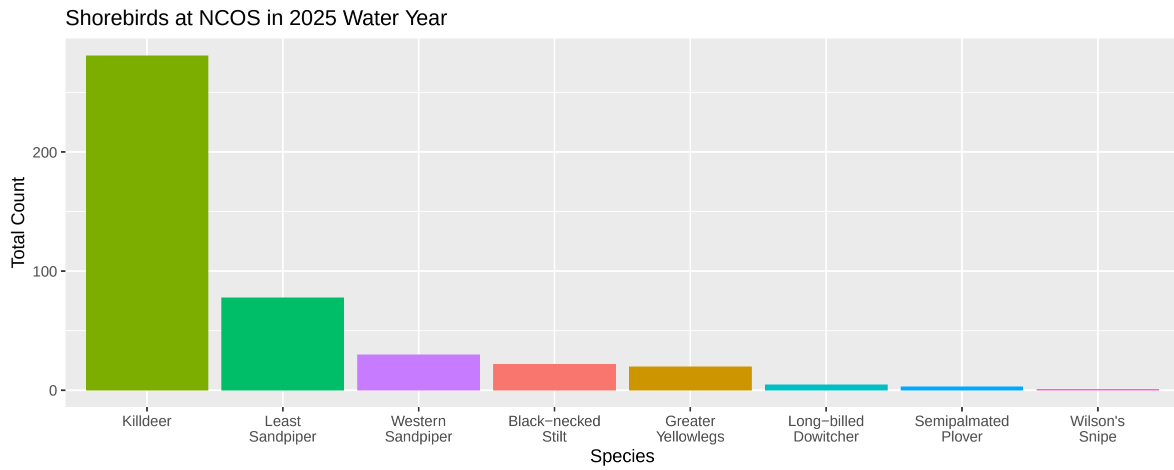
Wrangling Shorebird and Waterfowl

```
shorebird_waterfowl_25 <- bird_wy2025 |>
  #filter to only include herons, ibis and allies
  filter(e_bird_group %in% c("Waterfowl", "Shorebirds")) |>
  #group by species
  group_by(e_bird_group, observation_date) |>
  #calc total counts per species
  summarize(total_count = sum(count)) |>
  ungroup() |>
  left_join(venoco_daily,
  # naming shared columns between both data frames
  by = c("observation_date" = "date"))
```

3. Exploring Visualizations

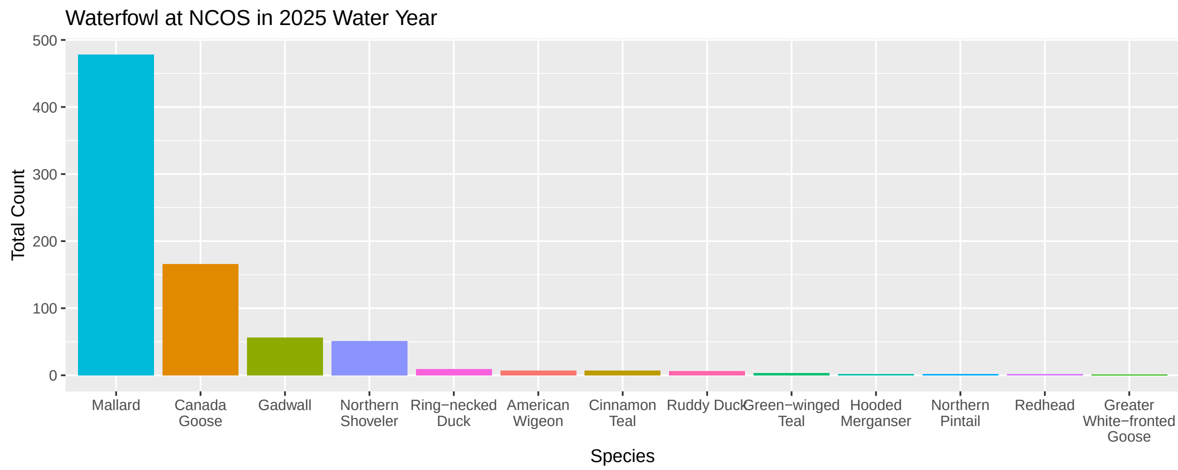
Plotting Shorebirds

```
ggplot(data = shorebird_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Shorebirds at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



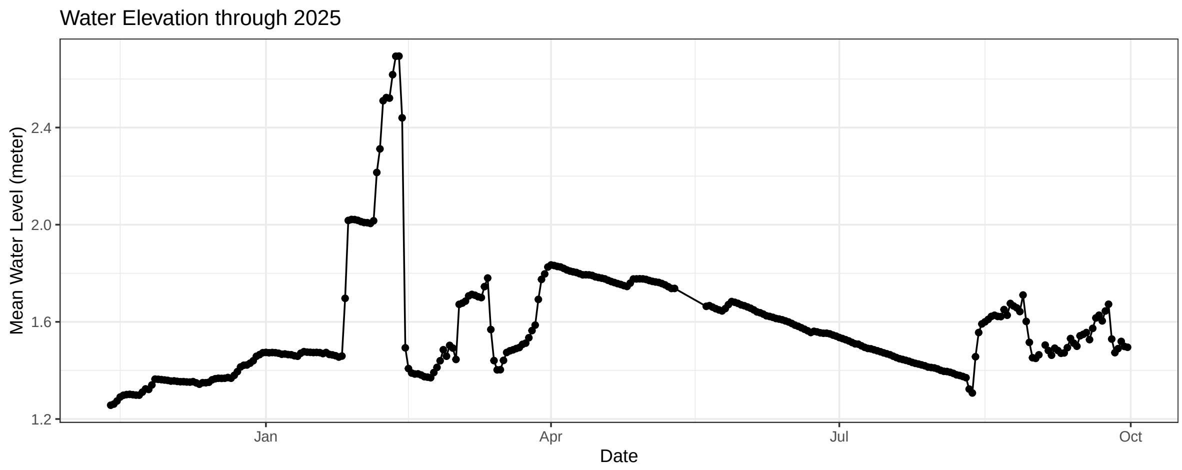
Plotting Waterfowl

```
ggplot(data = waterfowl_total_count,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = reorder(species, -total_count),
                     y = total_count,
                     fill = species)) +
  #first layer: column to represent counts
  geom_col() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Species",
       y = "Total Count",
       title = "Waterfowl at NCOS in 2025 Water Year") +
  scale_x_discrete(labels = ~ str_wrap(., width = 10))
```



Plotting Water Elevation of Time 2025

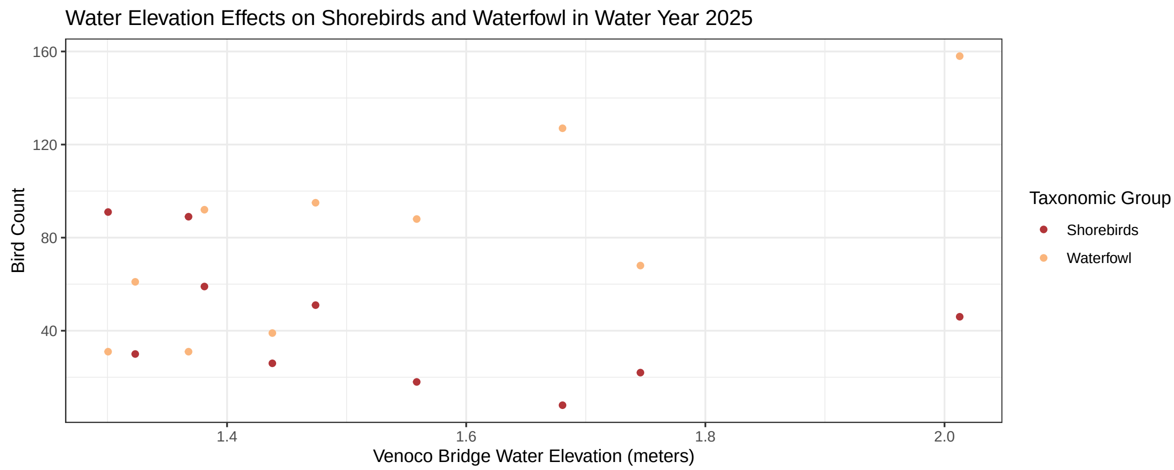
```
ggplot(data = venoco_daily,
       # x axis is species in descending order of total count
       # # y axis is total count
       # filling columns by species
       mapping = aes(x = date,
                     y = mean_level_m)) +
  #first layer: column to represent counts
  geom_point() +
  # adding line
  geom_line() +
  # custom theme
  theme_bw() +
  # taking out the legend
  theme(legend.position = "none") +
  #relabelling x and y axes, adding title
  labs(x = "Date",
       y = "Mean Water Level (meter)",
       title = "Water Elevation through 2025")
```




```
#scale_x_discrete(labels = ~ str_wrap(., width = 10))
```

Plotting Shorebirds and Waterfowl counts over different Water Elevations

```
ggplot(data = shorebird_waterfowl_25,
       aes(x = mean_level_m,
           y = total_count,
           color = e_bird_group)) +
  # setting point geometry
  geom_point() +
  # changing background theme
  theme_bw() +
  # custom colors
  scale_color_paletteer_d("nationalparkcolors::DeathValley") +
  # Adding axis titles, title and legend
  labs(
    x = "Venoco Bridge Water Elevation (meters)",
    y = "Bird Count",
    color = "Taxonomic Group",
    title = "Water Elevation Effects on Shorebirds and Waterfowl in Water
     ↪  Year 2025")
```



Addressing Rubric Feedback: water elevation is plotted with abundance using a scatter plot rather than a bar chart. Additionally, we are now comparing our response variable across bird groups instead of isolating their patterns.

Elective Planning

Idea: Illustrated informational bird species pamphlet for bird watchers, with insights on seasonal species abundance trends and ecological significance.

Drafts description/image

ElectivePlanPDF

Plan for making progress in the next two weeks

- Gather resources for information on most abundant species within the shorebird and Waterfowl taxonomic groups
  - KilldeerPDF
  - LeastSandpiperPDF
- Draft small ‘blurb’ descriptions for each species to accompany illustrations
- Complete illustrations
- Compile illustrations and written text into formatted pamphlet to be printed